

Laird Stewart

hlairdstew@gmail.com lairdstewart.com



EDUCATION

M.S. Computer Science, New York University 8/26–5/28

Johns Hopkins Engineering for Professionals (non-degree) 1/24–12/25

Theory of Machine Learning, Monte Carlo Methods, Electromagnetics, Fluid Dynamics, Plasma Physics

B.S. Computer Engineering, Georgia Institute of Technology 8/21–5/23*

4.0 GPA, Tau Beta Pi (Engineering honor society), Eta Kappa Nu (IEEE honor society)

B.S. Applied Math and Stats, Emory University 8/18–5/21*

3.9 GPA, Minor in Economics, Omicron Delta Epsilon (Economics Honor Society)

WORK EXPERIENCE

Software Engineer, Metron 6/23–7/26

Metron builds software grounded in applied math research for the Defense and Intelligence Community

- Core developer on TRK, the U.S. Navy's premier data fusion and tracking tool for anti-submarine warfare
- Implemented forward-backward particle smoothing feature in TRK, from algorithm to data visualization
- Lead engineer on ONR project to jointly estimate target submarine and environment state using acoustic-ray tracing. Leveraged normalizing flows and "ensemble samplers with affine invariance" for performance
- Organized a particle filter literature review group and a book club for Bishop's *Deep Learning* textbook

Data Science Intern, Equifax 5/21–8/21

- Built and automated end-to-end training pipeline for random forest classifier of U.S. households in R/SQL

Data Science Consultant, ERASE Child Trafficking 2/21–3/21

- Placed first in an Emory data science competition to help grow ERASE's Facebook outreach
- Built Linear Regression and Random Forest models on 2500 posts in Python to predict successful posts. Presented a 10-minute summary to ERASE leadership on how to increase engagement

Wealth and Investment Management Intern, Wells Fargo 7/20–8/20

- Researched six public equities and built discounted cash flow valuations to support portfolio composition

Tutor, Emory University 1/19–2/20

- Tutored Emory undergraduates in linear algebra and logic (philosophy) in 50 minute, 1-on-1 sessions

PUBLICATION

SIR Model of Time-Dependent Drug and Vaccine Distribution on COVID-19 4/21

Published in SIAM Undergraduate Research Online

- Expanded a standard "Susceptible-Infected-Recovered" (SIR) system of differential equations to include the effects of drug/vaccine deployment and modeled them in Python. Led team of four undergraduates.

ADDITIONAL INFORMATION

Skills: Java, Python, Linux, OpenGL, Protobuf/gRPC, Guice, Concurrent/Parallel programming in Java

Hobbies: Tennis, Squash, Running, Rock Climbing, Skiing, Backpacking, Hiking: summited 12 14,000 ft peaks

* Through Emory's Dual Degree Engineering Program, earned separate bachelor's degrees from Emory and Georgia Tech. Completed Emory degree requirements in three years (8/18–5/21) then transferred to Tech to complete engineering degree using transfer credits from Emory. Both degrees conferred 5/23.